

A Statistical Analysis of Air Pollution Levels Among High-Income Countries since 1990

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Summary

The goal of this project is to investigate trends in PM2.5 exposure and identify differences in air pollution levels across high-income countries and regions. Although high-income countries generally have more financial resources to address environmental problems, this study examines whether those resources are reflected in lower or improving PM2.5 exposure levels. Using statistical analysis and data visualizations, we analyzed PM2.5 exposure among 36 high-income countries from 1990 to 2023.

Overall, PM2.5 exposure decreased substantially among high-income countries during the study period. Of the 36 countries analyzed, 35 countries experienced a reduction in PM2.5 exposure, while only the Republic of Korea showed a slight increase. The mean percentage change was -38.36% , and the median percentage change was -42.86% , showing that decreases were widespread. Regional differences were also observed, with the Middle East showing the highest overall PM2.5 exposure and greatest variability, while North America had the lowest and most stable exposure levels. These findings suggest that high-income countries have generally improved air quality over time, but pollution outcomes still vary by country and region. This indicates that income alone does not fully explain PM2.5 exposure, and factors such as geography, industrial activity, energy use, and environmental policy may also play important roles.

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Study Description

Air pollution remains one of the most pressing sustainability challenges of the modern era, affecting billions of people across the globe regardless of national income or development status. In 2019, the World Health Organization estimated that approximately 99% of the world's population lived in areas where air quality failed to meet recommended guidelines. Fine particulate matter, known as PM_{2.5}, refers to airborne particles smaller than 2.5 microns in diameter that can penetrate deep into the respiratory system and enter the bloodstream, and has been linked to serious health conditions including lung disease, cardiovascular illness, and premature death.

Despite its association with industrialization and developing economies, high-income countries continue to record significant levels of PM_{2.5} exposure, raising important questions about the effectiveness of wealth and policy in combating pollution. These nations possess the financial resources and technological capacity to fund large-scale environmental solutions, yet many still struggle to meet safe air quality thresholds. Understanding how pollution levels vary across these countries and how they have changed over time is therefore both scientifically and politically relevant.

In this project, we analyze a dataset of annual PM_{2.5} exposure measurements spanning hundreds of countries and multiple years, focusing specifically on high-income nations. Using statistical analysis and data visualization, we aim to identify global trends in air pollution, highlight regional differences, and determine which high-income countries face the greatest exposure levels despite having the means to address them.

Data Explanation

The data used in this study comes from the State of Global Air database and includes annual PM_{2.5} exposure levels from 1990 to 2023. PM_{2.5} refers to fine particulate matter in the air that is 2.5 micrometers or smaller and is measured in micrograms per cubic meter, or $\mu\text{g}/\text{m}^3$.

Although the original dataset includes many countries, this study focuses only on 36 high-income countries. The main variables used were country, year, region, and mean PM_{2.5} exposure. The data also included lower and upper uncertainty estimates, but the analysis focused mainly on the mean PM_{2.5} values to compare pollution levels across countries and over time.

The dataset is useful because it provides repeated yearly measurements for each country, allowing us to examine long-term trends, regional differences, and country-level changes in air pollution. To measure improvement, we calculated the percentage change in PM_{2.5} exposure from 1990 to 2023 for each country. Negative percentage changes show decreases in pollution, while positive values show increases.

Overall, this data allows us to compare PM_{2.5} exposure among high-income countries and determine whether air pollution levels have improved or worsened over the past three decades.

Statistical Analysis and Findings

Time Trend

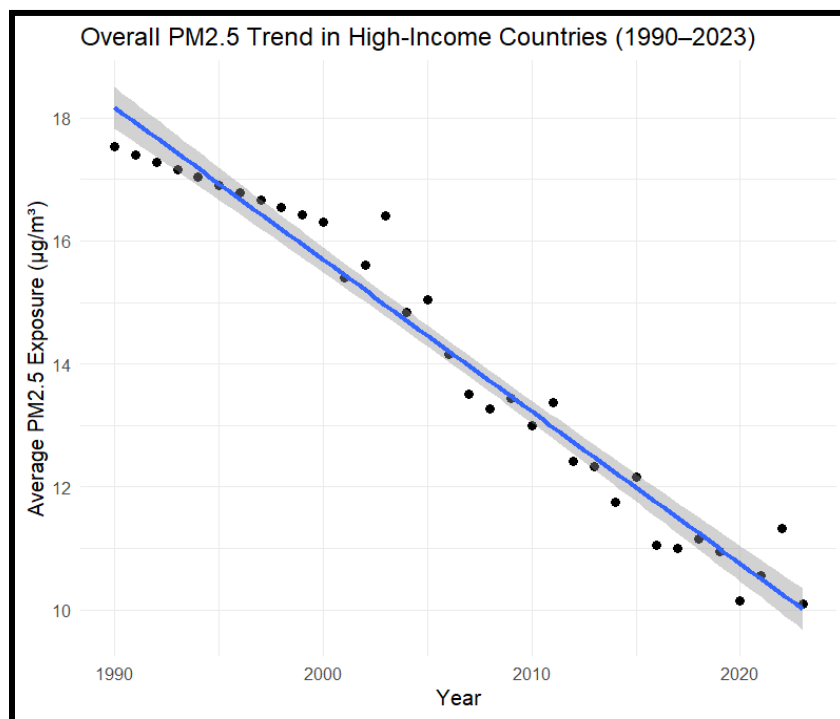


Figure 1. Overall PM2.5 Trend in High-Income Countries (1990–2023)

Figure 1 illustrates the average PM2.5 exposure across high-income countries over time. The data shows a steady decline from approximately 18 µg/m³ in 1990 to around 10 µg/m³ in 2023. The fitted regression line further confirms a strong negative trend, indicating that pollution levels have consistently decreased over time. Although minor fluctuations are observed throughout the period, the overall pattern demonstrates sustained improvement in air quality rather than short-term variation.

Trend by Region

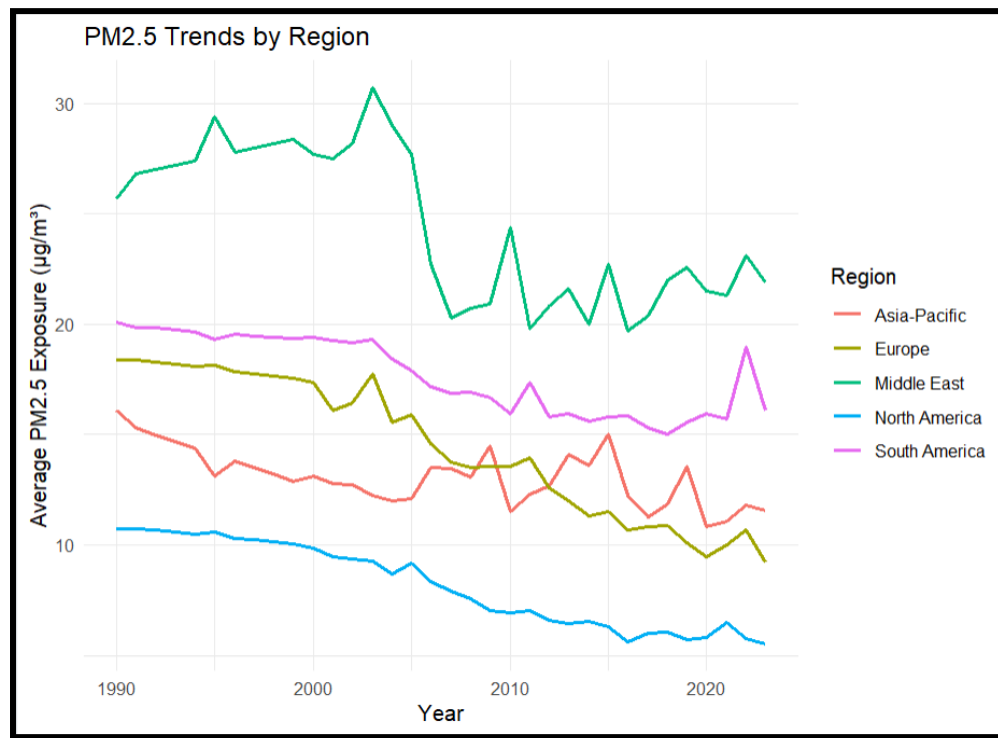


Figure 2. *PM2.5 Trends by Region*

Figure 2 compares PM2.5 exposure trends across major world regions. We observed multiple clear differences observed among regions. First, the Middle East consistently exhibits the highest pollution levels throughout the time period. Second, North America maintains the lowest PM2.5 levels across all years. Third, Europe and Asia-Pacific show moderate pollution levels with noticeable declines over time. Finally, South America displays moderate variability with a slight downward trend.

These results suggest that while all regions have generally improved, the magnitude and consistency of improvement vary significantly, likely due to differences in environmental policies, industrial activity, and geographic factors.

Percentage Change in Exposure by Country

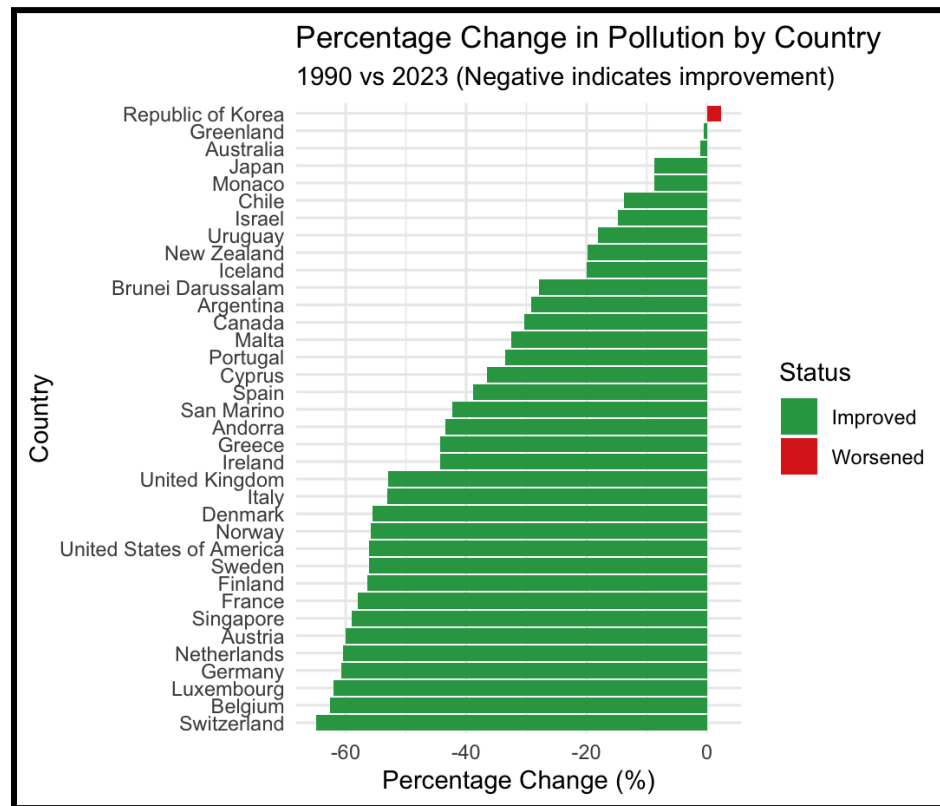


Figure 3. *Percentage Change in PM_{2.5} Exposure by Country (1990–2023)*

Figure 4 presents the percentage change in PM_{2.5} exposure for each country between 1990 and 2023. The majority of countries show negative values, indicating improvements in air quality. Switzerland experienced the largest reduction (−64.90%) and the Republic of Korea was the only country to show a slight increase (+2.34%).

Hypothesis Testing

To determine whether the observed decrease in PM_{2.5} exposure is statistically significant, a paired sample t-test was conducted comparing exposure levels in 1990 and 2023. The null hypothesis is that the mean difference in PM_{2.5} value in 1990 and 2023 is equal to 0 µg/m³. The alternative hypothesis is that the mean difference in PM_{2.5} value in 1990 and 2023 is not equal to 0 µg/m³.

$$H_0: \mu_{1990} - \mu_{2023} = \mu_d = 0$$

$$H_1: \mu_{1990} - \mu_{2023} = \mu_d \neq 0$$

The test statistic is $Z_0 = 8.43$ with 36 degrees of freedom. Since 8.43 is much larger than the critical value, 1.96, the result is significant. The p-value is 4.797×10^{-10} which is less than the alpha value of 0.05 so, we can reject the H_0 . The mean difference is 7.44 µg/m³ and the 95% confidence interval is [5.65, 9.23]. Since this entire range is above zero, it confirms a statistically significant decrease in pollution. This provides strong statistical evidence that PM_{2.5} exposure has significantly decreased over time among high-income countries.

Distribution of Switzerland versus Republic of Korea

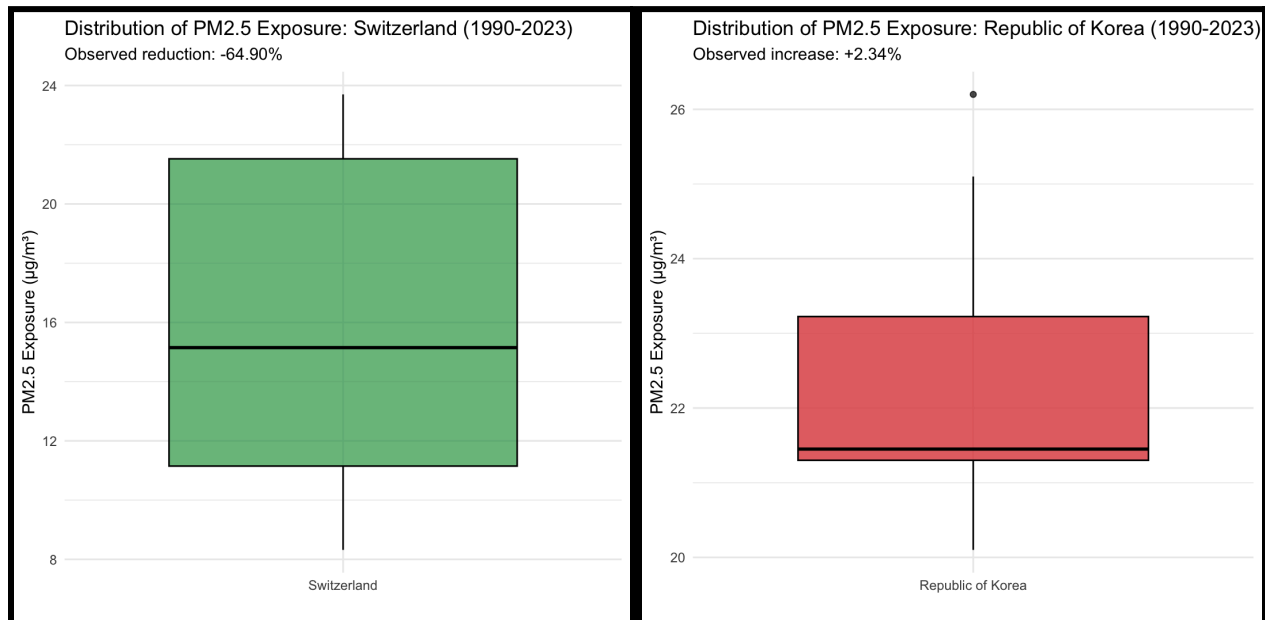


Figure 4. *Distribution of PM2.5 Exposure Switzerland and Republic of Korea Boxplot*

In figure 5, the box plot for Switzerland displays a large range and high variability. This tall box reflects a consistent and dramatic downward shift in pollution levels, moving from a median exposure of approximately 15 $\mu\text{g}/\text{m}^3$ with a maximum near 24 $\mu\text{g}/\text{m}^3$ down toward 8 $\mu\text{g}/\text{m}^3$. On the other hand, the Republic of Korea's box plot is highly compressed, indicating low variability. The majority of the annual observations are clustered tightly between 21 and 23 $\mu\text{g}/\text{m}^3$, with an outlier reaching above 26 $\mu\text{g}/\text{m}^3$. This stagnation confirms that while other high-income nations improved, Korea's air quality profile remained relatively unchanged. This visualization reinforces the conclusion that reductions in pollution are widespread among high-income countries, with only minimal exceptions.

Analysis using MSE

The purpose of this analysis is to examine differences in PM_{2.5} exposure across high-income countries and determine whether these differences are statistically meaningful. While earlier sections of this study identified general trends and improvements over time, this portion of the analysis focuses specifically on quantifying how much PM_{2.5} levels vary between countries. In particular, we aim to determine whether a single overall average can adequately represent pollution levels, or whether country-specific differences must be taken into account.

To accomplish this, we use the statistical concept of mean squared error (MSE) as a measure of variation and model accuracy. MSE evaluates how well a model's predicted values match the actual observed data by calculating the average of the squared differences between them. A larger MSE indicates that the model's predictions are far from the observed values, suggesting high variability or poor model fit. On the other hand, a smaller MSE indicates that the predictions closely match the data, meaning the model better represents the underlying pattern.

For this analysis, we compared two different models. The first model, the Overall Mean Model, assumes that all countries share the same PM_{2.5} exposure level, using a single global average to represent every observation. It does not account for differences between countries. The second model, the Country Mean Model, assigns each country its own average PM_{2.5} exposure level, allowing for variation between countries while still summarizing trends over time. By comparing the MSE values from these two models, we can evaluate whether accounting for country-level differences significantly improves the accuracy of the model.

The calculated MSE values are as follows: Overall Mean Model MSE: 40.06 and the Country Mean Model MSE: 10.01.

This represents a reduction of approximately 75% from the first model to the second. The substantial reduction in MSE indicates that the country mean model provides a significantly better fit to the data than the overall mean model. In other words, using a single global average results in large prediction errors, while accounting for country-specific averages greatly improves accuracy.

This finding demonstrates that PM_{2.5} exposure is not uniform across high-income countries. Instead, there is considerable variation between countries that cannot be captured by a single average value. The overall mean model performs poorly because it assumes similarity across all countries, an assumption that is clearly contradicted by the data. In contrast, the country mean model reduces error by incorporating differences between countries. This suggests that country-level factors play a critical role in determining air pollution levels.

Overall, the MSE analysis provides strong evidence that country-specific variation is a key component in understanding PM_{2.5} exposure patterns. The dramatic decrease in error when moving from a global model to a country-based model confirms that pollution levels differ significantly across high-income nations. Therefore, any meaningful analysis or policy approach should account for these differences rather than relying on a single generalized estimate.

Conclusion

Statement of Subject Matter Implications

The results of this study suggest that high-income countries have generally made meaningful progress in reducing PM2.5 exposure, but they also show that national income alone does not fully determine air quality outcomes. Because 35 of the 36 countries analyzed experienced decreases in PM2.5 exposure from 1990 to 2023, the findings indicate that many high-income countries have likely benefited from stronger environmental regulations, cleaner technologies, improved energy systems, and greater resources for pollution control. This is important because PM2.5 exposure is closely connected to public health risks, including respiratory and cardiovascular disease, so reductions in exposure represent a meaningful improvement for population health.

At the same time, the differences between countries and regions show that progress has not been equal. Some countries, such as Switzerland, experienced large reductions in PM2.5 exposure, while the Republic of Korea was the only country in the dataset to show a slight increase. Regional differences were also present, with the Middle East showing higher and more variable exposure levels and North America showing lower and more stable levels. These patterns suggest that factors such as industrial activity, population density, transportation systems, geography, climate, and policy enforcement may strongly influence air pollution outcomes even among countries with similar income levels.

The findings also imply that air pollution should not be viewed as a problem that automatically disappears once a country becomes wealthy. While high-income countries may

have more resources to invest in cleaner energy, monitoring systems, and pollution-reduction policies, those resources must be actively used to produce measurable improvements. Countries with persistent or higher PM_{2.5} exposure may need more targeted strategies, such as stricter emissions standards, cleaner transportation options, reduced reliance on fossil fuels, and policies designed for their specific geographic and industrial conditions.

Overall, this study shows that high-income countries have made broad progress in reducing PM_{2.5} exposure over the past three decades, but the unevenness of that progress is important. The results suggest that wealth can support environmental improvement, but policy choices, regional conditions, and country-specific factors play a major role in determining how much improvement actually occurs. Therefore, future efforts to reduce PM_{2.5} exposure should focus not only on economic capacity, but also on targeted environmental planning and consistent policy implementation.

Discussion of Further Questions

Although this study shows that PM_{2.5} exposure generally decreased among high-income countries from 1990 to 2023, the results also raise additional questions about why some countries improved more than others. Future research could examine the specific economic, policy, and geographic factors that contributed to these differences. For example, countries with stronger environmental regulations, cleaner energy sources, or more favorable geographic conditions may have been able to reduce PM_{2.5} exposure more effectively than countries with heavier industry, dense urban populations, or less favorable air circulation.

Another area for future research is the prediction of future PM_{2.5} exposure levels. This study identified a clear overall downward trend, but future analyses could test whether linear or nonlinear regression models are better at forecasting pollution levels. This would help determine whether current improvements are likely to continue or whether some countries may experience slower progress or future increases.

Finally, future research could focus on countries with the greatest year-to-year variability in PM_{2.5} exposure. High variability may suggest instability in environmental conditions, inconsistent policy enforcement, or sensitivity to temporary factors such as wildfires, industrial activity, or weather patterns. Studying these patterns more closely could help explain not only which countries improved, but also how stable and sustainable those improvements are over time.

References

Viewable Excel Link: [Data.xlsx](#)

State of Global Air. (n.d.). Explore the Data.

[https://www.stateofglobalair.org/data/#/air/table?country=USA&pollutant=pm25
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&hideCountry=false](https://www.stateofglobalair.org/data/#/air/table?country=USA&pollutant=pm25&measure=death&deathMetric=number&geography=country®ion=superReg&subregions=High-income&outcome=exposure®ionToggle=1&globals=false&hideCountry=false)